

Jing Xu (许靖)

EDUCATION

Tsinghua University

2021.9 – 2026.6

Ph.D. in Computer Science at IIIS

- **Advisor:** Andrew Chi-Chih Yao
- **Dissertation:** Understanding Fine-Tuning Mechanisms in Large Language Models: An Optimization Perspective. Received outstanding dissertation award of THU.

Peking University

2017.9 – 2021.6

B.S. in Artificial Intelligence at EECS (Turing Class, with summa cum laude)

- **Overall GPA:** 3.87/4.00
- **Ranking:** 1/93
- **Advisor:** Liwei Wang

RESEARCH INTERESTS

My research lies at the intersection of theoretical and applied machine learning. On the theoretical side, I am interested in establishing provable guarantees for the generalization and optimization of machine learning algorithms. On the empirical side, I have hands-on experience with large-scale LLM pre-training and am committed to designing efficient optimization algorithms that improve scalability and performance in pre-training.

I also have in-depth practical experience in quantitative research and have interned at top-tier quantitative trading firms, including Citadel Securities and Jump Trading.

I have worked on topics including:

- Efficient and stable optimizers for LLM pre-training.
- Adaptation of LLMs, e.g., parameter-efficient fine-tuning and scalable model merging.
- Generalization guarantees, implicit bias, and corresponding empirical signals in machine learning.
- Upper and lower convergence bounds for optimization algorithms on structured problems.

INDUSTRY EXPERIENCE

Qwen (Alibaba)

2026 – present

Researcher, Pre-training Team

- Conducting research on LLM pre-training, exploring novel approaches in model architectures, training strategies, and scaling techniques.
- Selected for the Alibaba AliStar Top Talent Program.

Citadel Securities

2025.6 – 2025.9

Quantitative Research Intern

- Built LLM pipelines to extract signals and build alphas from text-based alternative datasets. Received return offer.

Moonshot AI

2025.2 – 2025.6

Machine Learning Intern at pre-training team

- Participated in developing efficient and stable optimization algorithms (e.g. MuonClip and other variants of Muon) for LLM. Contributed to Kimi K2 model.

Jump Trading

2024.6 – 2024.8

Quantitative Research Intern

- Conducted alpha analysis for China's stock market.
- Developed a volatility prediction model.
- Designed and implemented a transaction cost prediction model.

PUBLICATIONS & TECHNICAL REPORTS

(* denotes equal contribution)

1. **Kimi K2: Open Agentic Intelligence**
Kimi Team *et al.*
2. **NeuralDB: Scaling Knowledge Editing in LLMs to 100,000 Facts with Neural KV Database** (ICLR 2026)
Weizhi Fei, Hao Shi, **Jing Xu**, Jingchen Peng, Jiazheng Li, Jingzhao Zhang, Bo Bai, Wei Han, Zhenyuan Chen, Xueyan Niu
3. **Scalable Model Merging with Progressive Layer-wise Distillation** (ICML 2025)
Jing Xu, Jiazheng Li, Jingzhao Zhang
4. **Understanding Nonlinear Implicit Bias via Region Counts in Input Space** (ICML 2025)
Jingwei Li*, **Jing Xu***, Zifan Wang, Huishuai Zhang, Jingzhao Zhang
5. **Near-Optimal Methods for Convex Simple Bilevel Problems** (Neurips 2024)
Huaqing Zhang*, Lesi Chen*, **Jing Xu**, Jingzhao Zhang
6. **Random Masking Finds Winning Tickets for Parameter Efficient Fine-tuning** (ICML 2024)
Jing Xu, Jingzhao Zhang
7. **On Bilevel Optimization without Lower-level Strong Convexity** (COLT 2024)
Lesi Chen*, **Jing Xu***, Jingzhao Zhang
8. **Towards Data-Algorithm Dependent Generalization Analysis: a Case Study on Overparameterized Linear Regression** (NeurIPS 2023)
Jing Xu*, Jiaye Teng*, Yang Yuan, Andrew C. Yao
9. **Quantifying the Variability Collapse of Neural Networks** (ICML 2023)
Jing Xu*, Haoxiong Liu*
10. **Faster Gradient-Free Algorithms for Nonsmooth Nonconvex Stochastic Optimization** (ICML 2023)
Lesi Chen, **Jing Xu**, Luo Luo

HONORS & AWARDS

- Outstanding Doctoral Dissertation Award of THU (2026.6)
- Excellent Graduate of THU (2026.6)
- Ubiquant Scholarship (2024.9)
- IIS Scholarship (2022.9 & 2023.9)
- Toyota Scholarship (2023.9)
- Excellent Graduate of PKU (2021.6)
- John Hopcroft Scholarship (2020.9)
- Turing Class Scholarship (2019.9)
- Award for Academic Excellence at PKU (2018.9)
- May 4th Scholarship at PKU (2018.9)

ACADEMIC SERVICES

- Served as a reviewer of ICML2022, 2024, 2025, 2026, Neurips2023, 2024, 2025, ICLR2024, 2025, CVPR2024, AAAI2025, AISTATS 2025